

INTERNSHIP REPORT

ON

STUDY AND CONFIGURATION OF A QUADCOPTER DRONE FLIGHT CONTROL SYSTEM

Submitted to
Nexonix Reflexo Pvt. Ltd.

Submitted by
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B.Tech – Electrical Engineering
Academic Session: 2026–27

This report is prepared in accordance with the prescribed internship report template and presents the learning, practical activities, project work, results, skills and industrial exposure gained during the internship.

CERTIFICATE

This page is reserved for the official Internship Completion Certificate issued by Nexonix Reflexo Pvt. Ltd.

STUDENT DECLARATION

I, Birendra Pratap Gautam, student of B.Tech in Electrical Engineering, hereby declare that this internship report titled “Study and Configuration of a Quadcopter Drone Flight Control System” is prepared on the basis of the learning and practical work completed during my internship at Nexonix Reflexo Pvt. Ltd.

I further declare that the material presented in this report has been compiled for academic purposes and represents my understanding of the concepts, activities and project work undertaken during the internship. Wherever external technical references have been used, they have been acknowledged in the References section.

I understand that the report may be evaluated by the institute and the organization as a record of my internship learning and industrial exposure.

Name: Birendra Pratap Gautam

Branch: Electrical Engineering

Date: _____

Signature: _____

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Nexonix Reflexo Pvt. Ltd. for providing me the opportunity to undertake an internship in the field of drone technology. The internship provided an important bridge between academic concepts and practical engineering applications.

I am thankful to the company mentor and the technical team for their guidance throughout the internship. Their explanations of drone components, aerodynamics, propulsion systems, electronics, software architecture and flight-controller configuration helped me understand how different engineering subsystems work together in an unmanned aerial vehicle.

I also express my gratitude to my faculty mentor and the institute for their continuous support and for encouraging practical learning. I appreciate my teammates and fellow students for their cooperation, discussions and constructive feedback during the learning activities.

Finally, I am grateful to everyone who directly or indirectly contributed to the successful completion of this internship and the preparation of this report.

ABSTRACT

This internship at Nexonix Reflexo Pvt. Ltd. provided practical exposure to the fundamentals of drone technology and the working of a multirotor flight-control system. The internship was structured around four major learning areas: Drone Technology Fundamentals and Components; Drone Aerodynamics, Propulsion and Electronics; Drone Software Architecture; and Flight Controller Configuration. The objective was to understand how mechanical, electrical and software subsystems are integrated to achieve stable and controlled flight.

During the internship, I studied the major components of a quadcopter, including the frame, motors, electronic speed controllers, propellers, battery, power distribution, flight controller and sensors. The aerodynamics module helped me understand thrust generation, lift, drag, weight and the effect of propeller rotation on vehicle motion. The propulsion and electronics section connected these concepts with motor control, power management and signal communication.

The software architecture module introduced the role of firmware, sensor data processing, control loops, configuration parameters and communication interfaces. Particular attention was given to flight-controller configuration, calibration, safety checks and the relationship between controller settings and stable flight. The selected project, “Study and Configuration of a Quadcopter Drone Flight Control System,” consolidated these concepts into a systematic study of the complete flight-control chain.

The internship improved my technical understanding, troubleshooting ability, documentation skills and confidence in working with interdisciplinary engineering systems. It also provided industrial exposure to the importance of safety, systematic testing, teamwork and accurate configuration. The overall experience helped me relate electrical engineering concepts to modern unmanned aerial systems and created a foundation for further learning in embedded systems, control engineering, robotics and autonomous drones.

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Note: After opening the document in Microsoft Word, the Table of Contents can be replaced/updated using References → Table of Contents if required. Page numbers may change slightly after inserting the official certificate and project screenshots.

CHAPTER 1: INTERNSHIP OBJECTIVES

1.1 Technical Skills

The primary technical objective of the internship was to develop a working understanding of drone hardware, propulsion, electronics, software architecture and flight-controller configuration. The learning was approached as an integrated system rather than as isolated components.

- Identify the major components of a quadcopter and understand their individual functions.
- Understand the relationship between motor speed, propeller thrust and vehicle movement.
- Study basic aerodynamic forces such as lift, drag, weight and thrust.
- Understand the role of electronic speed controllers and power distribution in a multirotor.
- Study the function of sensors and the flight controller in maintaining stable flight.
- Understand the basic software architecture used in a drone flight-control system.
- Learn the importance of calibration, parameter configuration and pre-flight safety checks.

1.2 Practical Learning

The internship converted theoretical concepts into practical engineering understanding. Instead of studying motors, sensors and control algorithms separately, the internship emphasized how these elements interact. For example, a change in motor command ultimately affects propeller thrust, vehicle attitude and sensor measurements, which are then processed by the flight controller to generate corrective commands.

This system-level approach helped me understand the importance of correct assembly, wiring, configuration and testing. It also showed why troubleshooting a drone requires a structured process rather than changing multiple parameters simultaneously.

1.3 Team Collaboration

Drone systems involve mechanical, electrical and software aspects, so collaboration is important. During the internship, discussions and peer learning helped in understanding technical terminology, interpreting observations and comparing configuration approaches. I learned to communicate technical issues clearly and to document observations before proposing a solution.

1.4 Problem Solving

Problem-solving was developed through systematic analysis. When an expected response was not obtained, the approach was to identify the subsystem involved, verify basic connections and

assumptions, review configuration parameters, and then test the system again. This method reduced guesswork and strengthened analytical thinking.

CHAPTER 2: WEEKLY INTERNSHIP ACTIVITIES

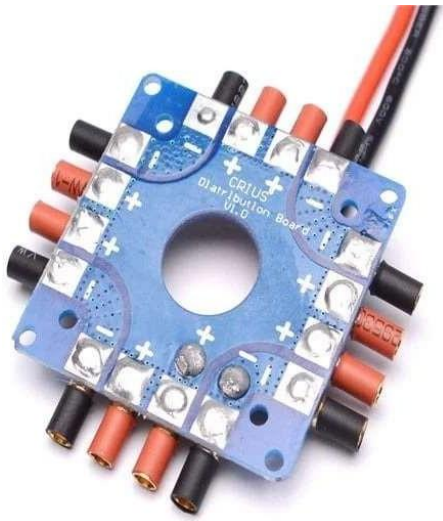
2.1 Week 1 – Drone Technology Fundamentals and Components

The first week focused on the fundamentals of unmanned aerial vehicles and the architecture of a multirotor drone. The learning began with the purpose and applications of drones, followed by the identification of the major hardware components used in a quadcopter.

Topic	Learning / Activity	Outcome
Drone basics	Studied UAV concept, multirotor configuration and common applications.	Understood the overall structure of a drone system.
Frame and structure	Examined the role of the frame and component placement.	Understood mechanical support and balance.
Motors and propellers	Studied motor-propeller combination and thrust generation.	Connected motor speed with propulsion.
Battery and power	Studied battery characteristics and power distribution.	Understood energy supply and electrical safety.
Flight controller	Studied controller, sensors and control role.	Understood the central role of the flight controller.

The main outcome of Week 1 was a clear component-level understanding of a quadcopter and the functional relationship between its electrical and mechanical subsystems.





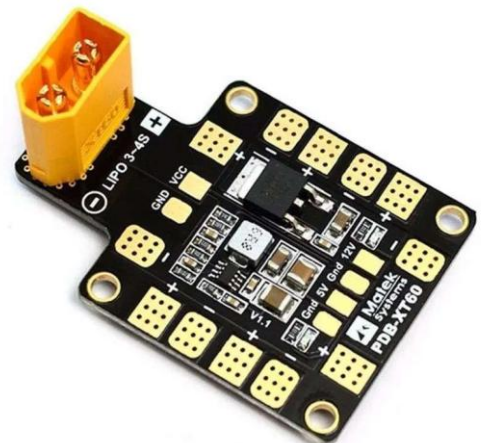
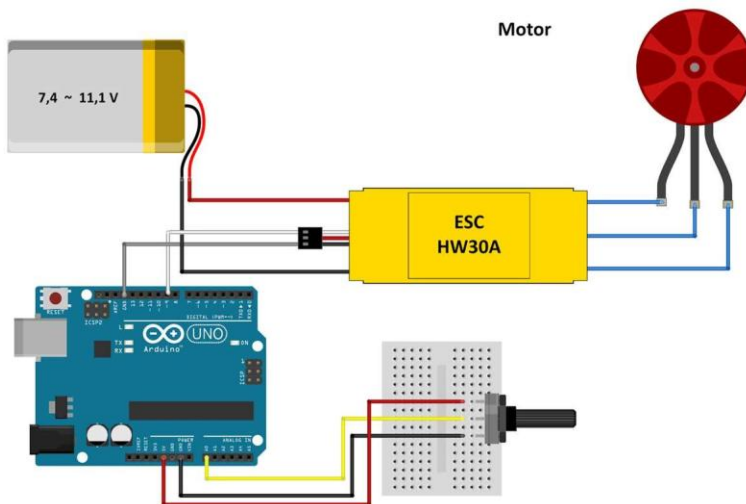
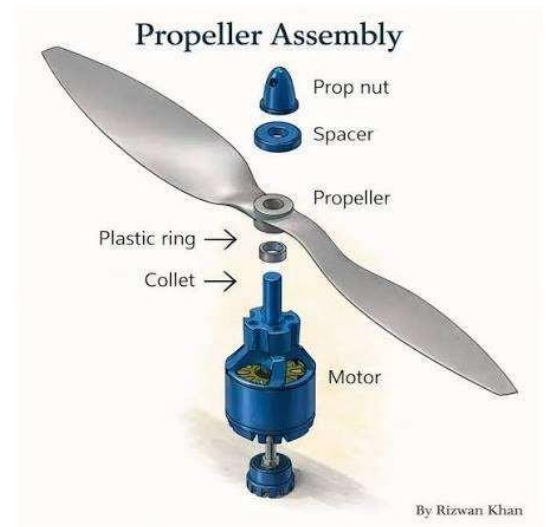
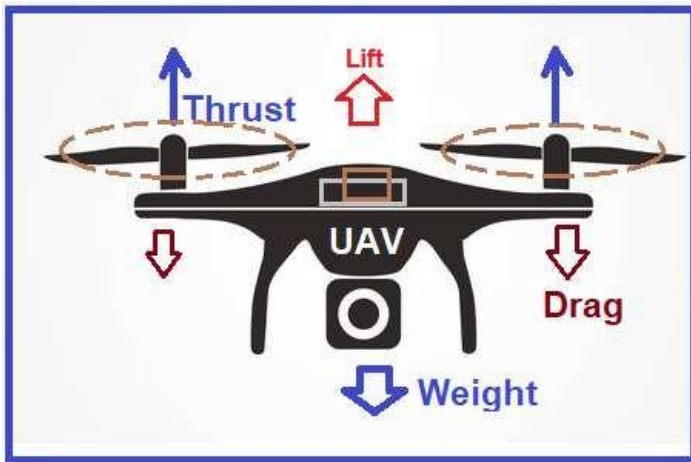
2.2 Week 2 – Aerodynamics, Propulsion and Electronics

The second week concentrated on the physical principles that enable a drone to fly. The concepts of thrust, lift, drag and weight were studied along with the effect of propeller rotation and motor speed on drone motion.

- Studied the four major forces acting on a flying vehicle: thrust, lift, drag and weight.
- Understood how propeller rotation produces thrust and how thrust is controlled by motor speed.
- Studied the function of electronic speed controllers in controlling brushless motors.
- Reviewed power distribution and the importance of correct voltage and current handling.
- Studied basic electrical connections between battery, power system, ESCs, motors and flight controller.
- Understood why correct motor order and propeller orientation are essential for stable operation.

2.3 Week 2 Practical Learning

The practical emphasis was on tracing the flow of electrical power and control signals through the drone. This helped clarify the difference between the high-power path used for propulsion and the lower-power signal path used for control and communication. The week also emphasized safe handling and checking before any powered test.



2.4 Week 3 – Drone Software Architecture

The third week introduced the software side of a drone. The flight controller was considered as an embedded computing unit that receives sensor information, processes it through control logic and generates commands for the motors.

Software Layer	Function	Importance
Firmware	Implements flight-control and system functions.	Provides the core behavior of the controller.
Sensor processing	Reads and interprets sensor measurements.	Provides attitude and motion information.
Control loop	Compares desired and measured states.	Generates corrective commands.
Configuration	Stores system and tuning parameters.	Adapts the controller to the vehicle.
Communication	Transfers data between controller and external tools.	Supports monitoring and configuration.

The major learning outcome was understanding that drone flight is a closed-loop control problem. The controller continuously uses measurements from sensors to determine whether the actual state of the drone matches the desired state and then adjusts the actuator commands.



Figure 2.11: Flight Controller Configuration Interface (Dashboard)

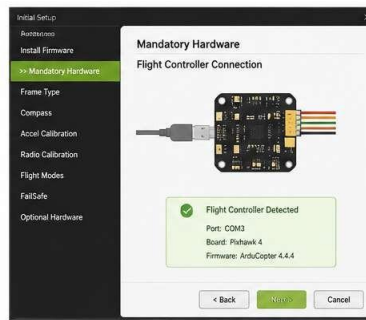


Figure 2.12: Flight Controller Connection Screen

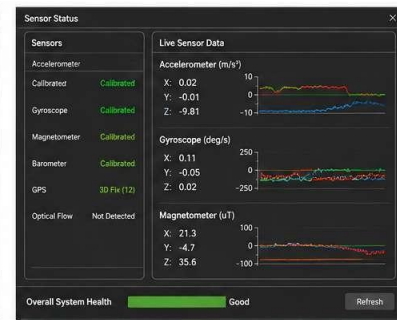


Figure 2.13: Sensor Status and Monitoring

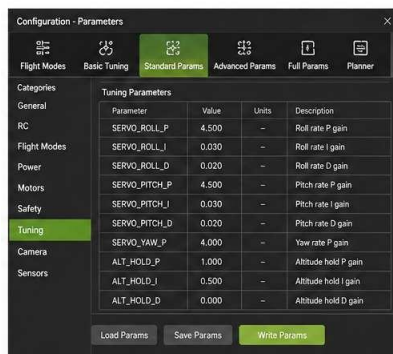


Figure 2.14: Flight Controller Parameters

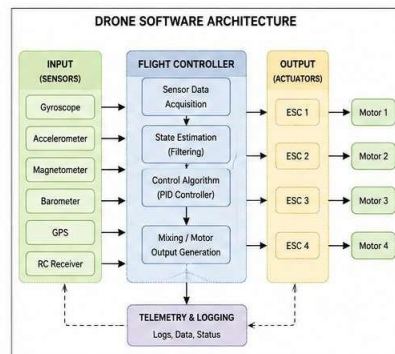


Figure 2.15: Drone Software Architecture

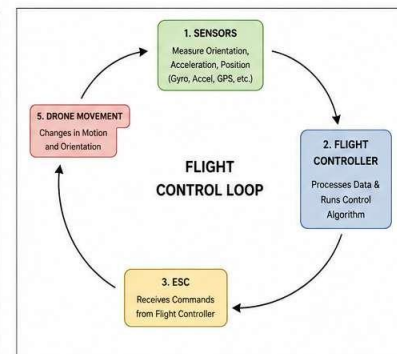


Figure 2.16: Control Loop of Flight Controller

2.5 Week 4 – Flight Controller Configuration

The fourth week focused on the configuration and calibration aspects of a flight controller. The activities emphasized systematic setup rather than random parameter changes.

- Reviewed the purpose of flight-controller configuration software.
- Studied sensor calibration and why accurate sensor readings are necessary.
- Reviewed frame type and motor-layout configuration.
- Studied radio/control input calibration and the importance of correct channel mapping.
- Reviewed safety-related configuration and pre-flight checks.
- Studied the effect of configuration parameters on system response and stability.
- Understood the importance of recording configuration changes and test observations.

The final week connected the hardware and software learning from the previous weeks. It demonstrated that even a correctly assembled drone can behave incorrectly if the flight controller is configured improperly. Therefore, calibration, parameter verification and testing are essential stages of system integration.

WEEK 4 – FLIGHT CONTROLLER CONFIGURATION

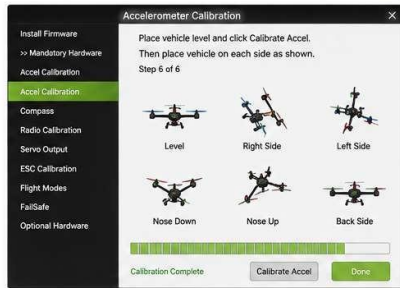


Figure 4.1: Accelerometer Calibration

Accelerometer calibration is performed by placing the drone in all six positions. This ensures accurate measurement of linear acceleration and stable flight control.

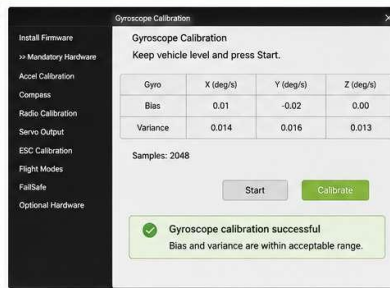


Figure 4.2: Gyroscope Calibration

Gyroscope calibration removes bias and drift from the sensors. The values are checked to be within the acceptable range for smooth and stable flight performance.

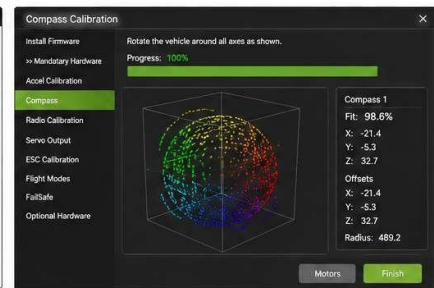


Figure 4.3: Compass Calibration

Compass calibration is done by rotating the drone in all directions. A good calibration shows a sphere shape with high fit percentage, ensuring accurate heading information.

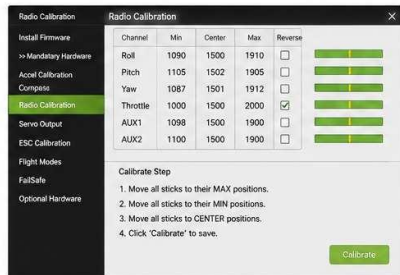


Figure 4.4: Radio (Transmitter) Calibration

Radio calibration maps the transmitter inputs to the flight controller channels. Correct min, max and center values ensure precise control of the drone.

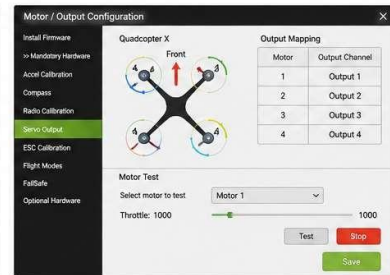


Figure 4.5: Motor / Output Configuration

Motor outputs are mapped to the correct channels. Motor test is performed to verify the direction and operation of each motor before arm and flight.

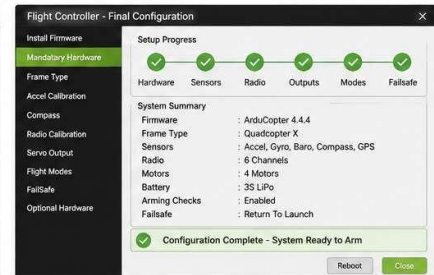


Figure 4.6: Final Flight Controller Configuration

Final configuration summary shows all modules are set. The flight controller is now ready for arming and safe flight operations.

CHAPTER 3: PROJECT INTRODUCTION

3.1 Project Title

Study and Configuration of a Quadcopter Drone Flight Control System

3.2 Introduction

A quadcopter is a multirotor unmanned aerial vehicle that uses four motor-propeller units to generate and control thrust. The flight controller acts as the central control unit and combines sensor measurements, pilot commands and control algorithms to regulate the motors. The project studied this complete chain from input to controlled response.

3.3 Objectives

- Understand the architecture of a quadcopter flight-control system.
- Identify the major hardware and software elements involved in flight control.
- Understand how sensor measurements are used for attitude stabilization.
- Study the configuration sequence required before a controlled test.
- Relate electrical engineering and control concepts to a practical UAV system.

- Develop a systematic approach to troubleshooting and configuration.

3.4 Scope

The project scope covers the study of a conventional quadcopter architecture, including propulsion, power, sensing, flight-control software and configuration. It focuses on the fundamentals of system integration and does not claim to cover every autonomous-navigation, computer-vision or advanced mission-planning feature used in commercial UAVs.

3.5 Applications

- Aerial photography and videography
- Surveying and mapping
- Inspection of infrastructure
- Agricultural monitoring
- Search and observation tasks
- Educational and research platforms
- Robotics and autonomous-system development

3.6 Working Principle of the Project

The working principle can be understood as a feedback-control loop. A desired command is provided to the flight controller either by the pilot or by a higher-level system. Sensors measure the current motion and orientation of the vehicle. The controller compares the measured state with the desired state and calculates corrective actions. These actions are converted into motor commands through the actuator-control system. The resulting motion changes the sensor readings, closing the feedback loop.

For a quadcopter, coordinated changes in the speed of the four motors produce vertical motion, roll, pitch and yaw. The controller must maintain the appropriate balance between these motor outputs to keep the vehicle stable.

3.7 System Flow

Stage	Input	Processing	Output
1	Pilot / command	Desired attitude or throttle interpreted	Target state
2	Sensors	Measure acceleration, angular motion and orientation	Measured state
3	Controller	Compare target and measured state	Correction signal
4	Motor control	Convert correction into actuator commands	Motor commands
5	Propulsion	Motor-propeller units generate thrust	Vehicle motion
6	Feedback	New motion measured by sensors	Updated state

3.8 Project Limitations

The project was primarily educational and configuration-oriented. Detailed autonomous navigation, advanced computer vision, long-range communication, payload optimization and commercial flight certification were outside the intended scope. Actual hardware specifications should be inserted from the equipment used during the internship wherever the institute requires exact component-level data.

CHAPTER 4: HARDWARE / SOFTWARE USED

4.1 Hardware

Hardware	Purpose	Remarks
Quadcopter frame	Mechanical support	Provides structural mounting.
Brushless motors	Generate propulsion	Motor speed determines thrust.
Propellers	Convert rotation into thrust	Correct size and orientation are essential.
Electronic Speed Controllers	Control motor speed	Receive control commands from the flight controller.
Battery	Provides electrical energy	Must be compatible with system requirements.
Power distribution	Distributes electrical power	Supports propulsion and electronics supply.
Flight controller	Runs control logic	Processes sensor data and produces motor commands.
IMU / sensors	Measure motion and orientation	Provides feedback for stabilization.
Radio receiver	Receives pilot commands	Transfers control inputs to the controller.

4.2 Hardware Integration

The hardware is integrated so that the battery supplies the propulsion and electronics power paths, the flight controller receives sensor and command inputs, and the ESCs regulate the motors. Correct wiring, connector placement, motor order and propeller direction are essential. A configuration error in any of these areas can result in an incorrect or unstable response.

4.3 Software

Software / Category	Purpose
Flight-controller firmware	Executes control and stabilization functions.
Configuration software / ground-control interface	Used for calibration, parameter configuration and monitoring.
Operating system utilities	Support communication, file handling and documentation.
Documentation tools	Used for recording configuration and observations.

4.4 Programming / Control Concepts

Although the internship was primarily configuration-oriented, it involved several programming and control concepts. These included embedded firmware, sensor data processing, feedback control, actuator output, parameterization and communication protocols. Understanding these concepts is important because a flight controller is effectively a real-time embedded control system.

4.5 Libraries and Tools

Specific software libraries depend on the flight-controller platform and firmware used during practical work. The report therefore avoids assigning unsupported library names. The main tools involved in the learning process were flight-controller configuration software, serial/communication interfaces, documentation utilities and testing procedures.

CHAPTER 5: RESULTS AND DISCUSSION

5.1 Testing

Testing was approached in stages. First, the basic hardware and configuration assumptions were reviewed. Next, sensor and input calibration concepts were checked. Motor and actuator behavior was then considered in relation to the configured frame type and motor layout. Finally, the complete feedback-control chain was reviewed conceptually.

Test Item	Expected Result	Observation / Outcome
Component identification	All major components correctly identified	Achieved
Sensor calibration concept	Sensor readings suitable for control	Understood and reviewed
Input calibration	Commands mapped correctly	Understood and reviewed
Motor order/configuration	Each output corresponds to correct motor	Configuration principle understood
Propeller orientation	Correct thrust direction	Importance verified
Safety checks	System can be tested systematically	Pre-flight sequence established

5.2 Output

The major output of the project was a consolidated understanding of how a quadcopter flight-control system is configured and operated. The work produced a structured learning record covering hardware identification, propulsion, electronics, software architecture, calibration and system-level troubleshooting.

5.3 Results

- Developed a clear understanding of quadcopter architecture.
- Understood the relationship between motor commands and vehicle motion.
- Learned why sensor calibration is important for stable control.
- Understood the role of feedback in flight stabilization.
- Developed a systematic approach to configuration and troubleshooting.
- Improved technical documentation and engineering communication.

5.4 Graphs and Analysis

A useful way to analyze a flight-control system is to study how the commanded response changes with time and how quickly the actual response follows it. In a practical test, parameters such as attitude angle, angular rate, throttle command and motor output can be logged and plotted. The exact numerical graphs should be generated from the actual flight-controller log files available to the student.

Suggested Graph	X-axis	Y-axis	Purpose
Desired vs. actual attitude	Time	Angle	Evaluate tracking and stabilization.
Angular rate	Time	Angular rate	Observe response and damping.
Throttle / motor output	Time	Command level	Study actuator behavior.
Battery voltage	Time	Voltage	Observe power-system behavior.

5.5 Challenges

- Understanding the interaction between mechanical, electrical and software subsystems.
- Learning technical terminology related to UAV propulsion and flight control.
- Understanding why calibration and configuration must be completed in a fixed sequence.
- Connecting sensor feedback with the resulting motor response.
- Maintaining a structured troubleshooting process instead of changing several parameters simultaneously.

5.6 Future Improvements

- Integrate autonomous waypoint navigation.
- Add GPS-based position control and telemetry analysis.
- Explore computer-vision-assisted navigation.
- Study advanced PID tuning and controller optimization.
- Develop automated pre-flight diagnostic checks.
- Analyze real flight logs for quantitative performance evaluation.

CHAPTER 6: SKILLS LEARNED

6.1 Technical Skills

- Drone component identification and system architecture
- Basic UAV aerodynamics and propulsion
- Brushless motor and ESC concepts
- Power distribution and electrical safety awareness
- Flight-controller architecture and sensor feedback
- Calibration and configuration principles
- Basic feedback-control understanding
- Technical documentation and system analysis

6.2 Soft Skills

The internship also improved soft skills such as discipline, time management, observation, patience and responsibility. Working with a technical system taught me that small errors in documentation or configuration can affect the entire system, so careful work is essential.

6.3 Communication

I learned to communicate technical observations using clear terminology and logical sequences. Instead of simply stating that a system was not responding correctly, I learned to describe the observed behavior, identify the likely subsystem and explain the checks performed. This approach is useful in both academic projects and professional engineering environments.

6.4 Leadership

The internship provided opportunities to take ownership of assigned learning tasks, organize notes and help coordinate discussions. Leadership in a technical environment is not limited to giving instructions; it also involves listening, documenting decisions, supporting team members and ensuring that tasks are completed systematically.

6.5 Industrial Exposure

The internship provided exposure to an engineering workflow in which multiple disciplines are integrated into a single product. A drone combines mechanical structure, electrical power, electronics, embedded software, control theory and communication. This experience helped me understand why industrial development requires coordination between different technical roles.

Another important lesson was the importance of safety and verification. Before a system is tested, basic assumptions must be checked and the configuration must be reviewed. This mindset is directly applicable to electrical engineering laboratories, embedded systems and industrial control applications.

6.6 Skill Development Summary

Skill Area	Before Internship	After Internship
Drone technology	Basic awareness	System-level understanding
Aerodynamics	Theoretical familiarity	Applied understanding of thrust and forces
Flight controller	Limited knowledge	Understood configuration and feedback role
Troubleshooting	General approach	More systematic and evidence-based
Documentation	Academic reports	Technical internship documentation
Teamwork	Academic collaboration	Technical discussion and task coordination

CHAPTER 7: CONCLUSION

7.1 Summary

The internship at Nexonix Reflexo Pvt. Ltd. was a valuable practical learning experience in drone technology and flight-control systems. The four-week learning structure covered drone fundamentals and components, aerodynamics and propulsion, electronics, software architecture and flight-controller configuration. These topics collectively provided a system-level understanding of how a quadcopter is designed, controlled and configured.

The project on the study and configuration of a quadcopter flight-control system helped connect theoretical engineering concepts with a modern real-world application. The work demonstrated the importance of feedback, sensor calibration, motor control, correct configuration and systematic testing.

7.2 Learning Outcomes

- Ability to identify and explain major quadcopter components.
- Understanding of the basic forces and principles involved in multirotor flight.
- Understanding of propulsion and electronic control elements.
- Knowledge of the role of a flight controller and sensor feedback.
- Awareness of software architecture and configuration parameters.
- Improved troubleshooting, documentation and technical communication skills.
- Better understanding of industrial safety and systematic testing.

7.3 Final Conclusion

Overall, the internship strengthened my interest in interdisciplinary engineering systems and showed how electrical engineering principles are applied in modern unmanned aerial vehicles. The experience has provided a foundation for further study in embedded systems, control engineering, robotics, autonomous systems and advanced UAV technologies.

REFERENCES

8.1 IEEE Format

- [1] J. G. Leishman, Principles of Helicopter Aerodynamics, Cambridge University Press.
- [2] R. C. Dorf and R. H. Bishop, Modern Control Systems, Pearson.
- [3] A. K. Sawhney, A Course in Electrical and Electronic Measurements and Instrumentation, Dhanpat Rai.
- [4] Technical documentation and training material provided during the internship at Nexonix Reflexo Pvt. Ltd.

8.2 APA Format

- Dorf, R. C., & Bishop, R. H. (2017). Modern control systems. Pearson.
- Leishman, J. G. (2006). Principles of helicopter aerodynamics. Cambridge University Press.

8.3 Books

The books listed above were selected as supporting references for control systems, aerodynamics and electrical/electronic concepts relevant to the internship.

8.4 Websites / Online Documentation

Official flight-controller and UAV documentation may be cited here according to the exact platform used during the internship. The student should insert the exact documentation URLs and access dates corresponding to the hardware/firmware actually used.

8.5 Research Papers

For the final academic submission, research papers related to multirotor dynamics, PID control, UAV stabilization and autonomous navigation may be added after verifying the exact papers consulted.